

CO2 Assessment Tool 1

Software Requirement Specification (SRS)

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Project Title: Intelligent Nuclear Power Plant Monitoring Platform

1. Study and Analyze the Scenario

Business Domain

The project belongs to the Energy and Nuclear Power Management domain. It monitors nuclear power plant operations, improves safety, and supports efficient plant performance through continuous monitoring.

Stakeholders

- Plant Operator
- Maintenance Engineer
- Safety Officer
- Plant Administrator
- Regulatory Authority
- Plant Management

Existing Challenges

- Manual monitoring of reactor conditions.
- Delay in detecting equipment failures.
- Difficulty in tracking radiation levels continuously.
- High operational risk due to unexpected failures.

- Time-consuming compliance reporting.
- Delayed emergency response.

System Objectives

- Monitor reactor conditions.
- Detect operational anomalies.
- Predict equipment failures.
- Track radiation levels.
- Support emergency response.
- Generate compliance analytics.

2. Identify Functional and Non-Functional Requirements

Functional Requirements

- Monitor reactor temperature, pressure, coolant flow, and radiation levels.
- Collect real-time sensor data.
- Detect abnormal operating conditions.
- Generate emergency alerts.
- Predict equipment failures.
- Maintain operational logs.
- Generate compliance reports.
- Provide secure user login.
- Display live monitoring dashboards.
- Store historical operational data.
- Support maintenance scheduling.
- Generate analytical reports.

Non-Functional Requirements

Performance

- Process sensor data with minimum delay.
- Generate alerts immediately.

Security

- Allow only authorized users.

- Encrypt sensitive data.

Reliability

- Operate continuously with minimal downtime.
- Prevent data loss.

Usability

- Provide a simple dashboard.
- Display alerts clearly.

Scalability

- Support additional reactors and sensors.

Maintainability

- Allow easy software updates.

Availability

- Operate 24×7.

3. Identify Actors and Use Cases

Primary Actors

- Plant Operator
- Maintenance Engineer
- Safety Officer

Secondary Actors

- Administrator
- Regulatory Authority

Major Use Cases

- Plant Operator: Monitor reactor, view dashboard, receive alerts, generate reports.
- Maintenance Engineer: View equipment health, predict failures, schedule maintenance.
- Safety Officer: Monitor radiation, respond to alerts, verify compliance.
- Administrator: Manage users and system settings.

- Regulatory Authority: Review compliance reports.

Use Case Diagram: (Insert UML Use Case Diagram here.)

4. Prepare the Software Requirement Specification (SRS)

Introduction

The Intelligent Nuclear Power Plant Monitoring Platform continuously monitors reactor operations, detects anomalies, predicts failures, tracks radiation, supports emergency response, and generates compliance reports.

Problem Description

Manual monitoring may delay detection of failures and radiation leaks. The proposed system automates monitoring and improves safety and efficiency.

Scope of the System

Monitors reactor parameters, predicts failures, supports emergency response, stores records, and generates compliance reports.

Functional Requirements

Reactor monitoring, anomaly detection, failure prediction, radiation monitoring, emergency alerts, compliance reporting, dashboards, authentication, and historical data management.

Non-Functional Requirements

High performance, security, reliability, usability, scalability, maintainability, and availability.

Use Case Descriptions

UC1 Monitor Reactor Conditions; UC2 Detect Operational Anomalies; UC3 Predict Equipment Failures; UC4 Track Radiation Levels; UC5 Generate Compliance Reports.

Data Requirements

Stores reactor, sensor, radiation, equipment, user, maintenance, alert, and compliance data.

External Interface Requirements

User Interface: Login, Dashboard, Alerts, Reports. Hardware: Sensors. Software: Database, Analytics, Reporting. Communication: Secure network, Email/SMS.

System Features

Real-time monitoring, anomaly detection, failure prediction, radiation tracking, emergency support, compliance analytics, secure authentication, interactive dashboard.

5. Identify Assumptions and Constraints

Assumptions

- Sensors provide accurate data.
- Stable network connectivity is available.
- Users are trained.
- Continuous power supply is available.
- Database backup is performed regularly.

Constraints

Technical

- Depends on sensor accuracy.
- Requires reliable network.

Operational

- Requires trained operators.
- Continuous monitoring is necessary.

Economic

- High installation and maintenance cost.

Legal

- Must comply with nuclear safety regulations.

Time-related

- Alerts should be generated immediately.

- Reports should be generated on time.

6. Validate Requirement Completeness and Consistency

Requirement Completeness

All stakeholder, functional, and non-functional requirements are covered.

Consistency

Requirements do not conflict and support the project objectives.

Ambiguity Check

Requirements are clear and understandable.

Missing Requirements

No major requirements are missing.

Suggested Improvements

Add AI-based prediction, cloud backup, mobile application, improved dashboard, and multi-factor authentication.

Expected Outcomes

- Enhanced plant safety.
- Reduced operational risk.
- Improved regulatory compliance.
- Lower unplanned downtime.
- Better decision support.